

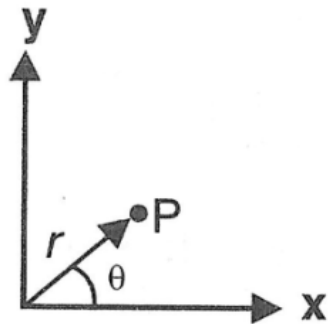
Chapter 1: Cartesian Coordinate Systems

Fletcher Dunn
Valve Software

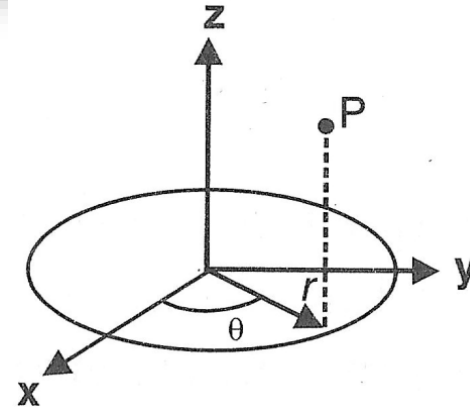
Ian Parberry
University of North Texas

3D Math Primer for Graphics and Game Development

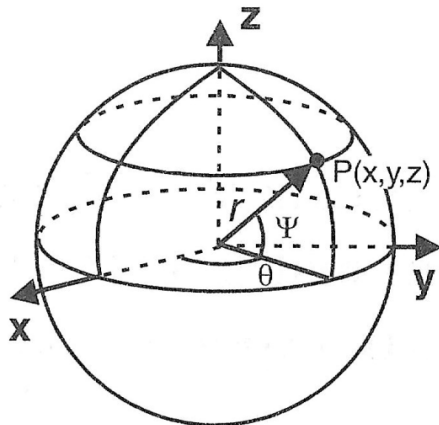
Sistemas de Coordenadas



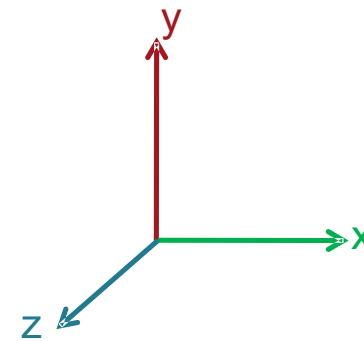
Coordenadas Polares



Coordenadas Cilíndricas



Coordenadas Esféricas



Coordenadas Cartesianas

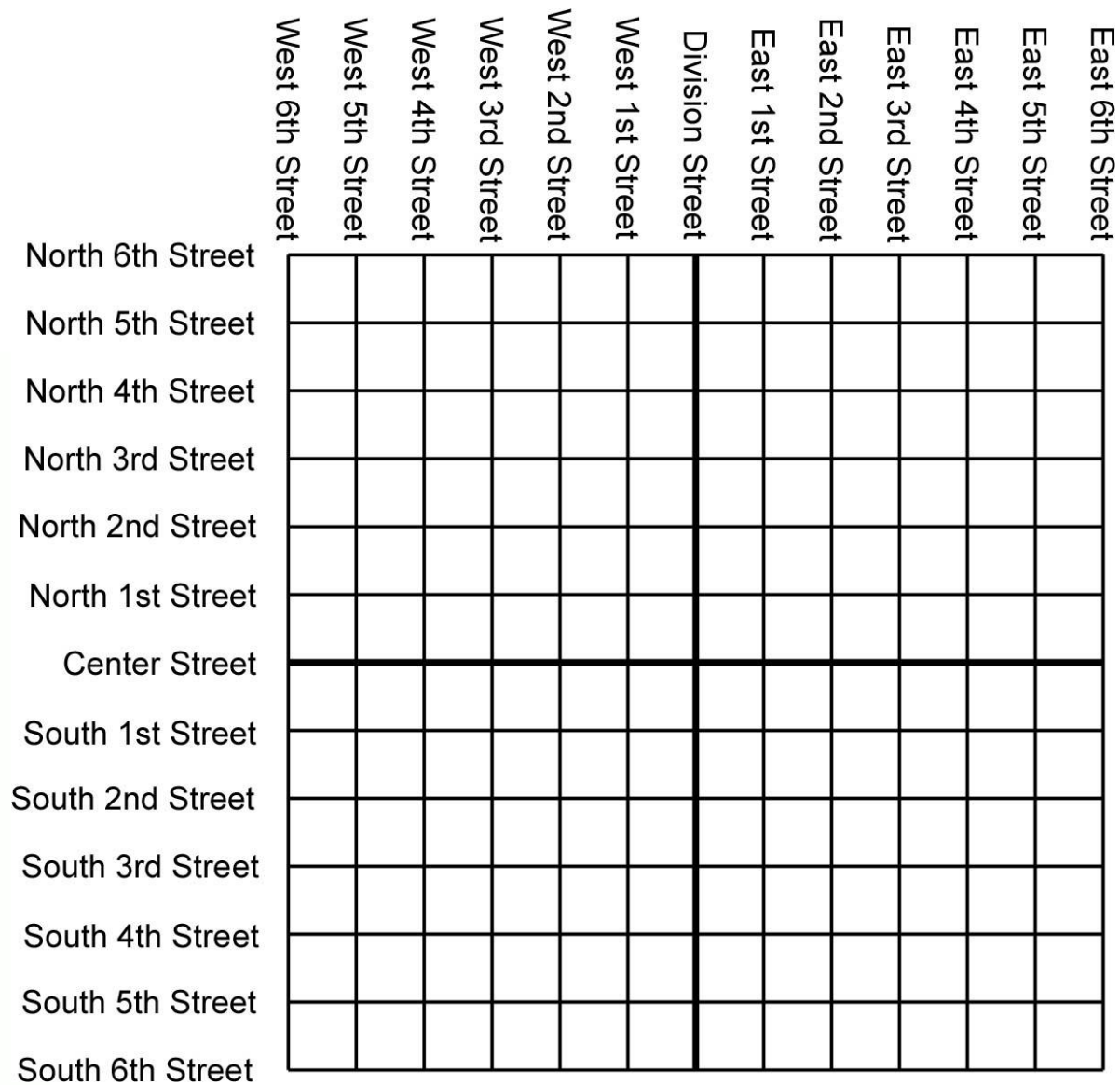


Introduction

- 3D math is all about measuring locations, distances, and angles precisely and mathematically in 3D space.
- The most frequently used framework to perform such calculations using a computer is called the *Cartesian coordinate system*.

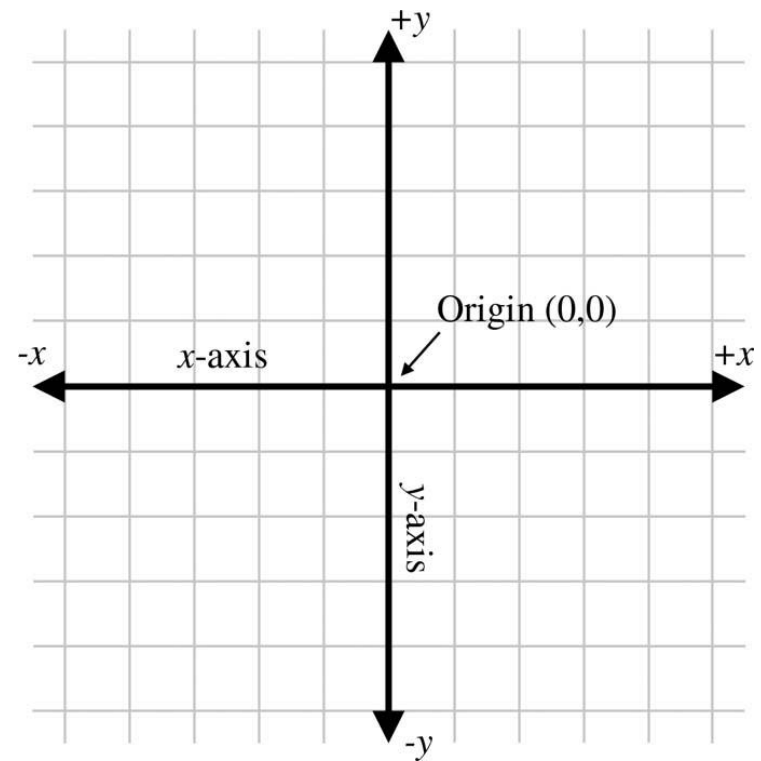
1D Mathematics

- We assume that you already know about the natural numbers, the integers, the rational numbers, and the real numbers.
- On a computer you have to make do with **shorts**, **ints**, **floats**, and **doubles**. These have limited precision.
- We assume that you have a basic understanding about how numbers are represented on a computer.
- Remember the First Law of Computer Graphics: *If it looks right, it is right.*



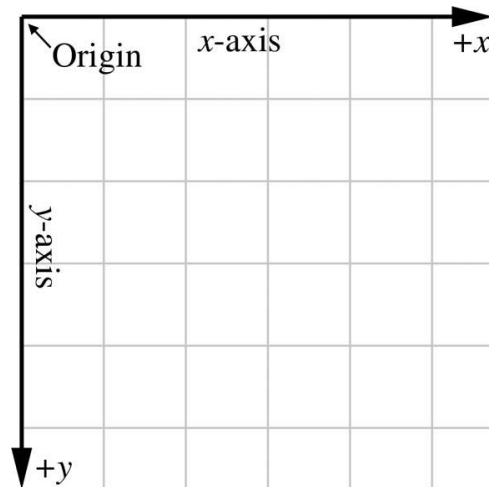
2D Coordinate Spaces

- All that really matters are the numbers.
- The abstract version of this is called a *2D Cartesian coordinate space*.



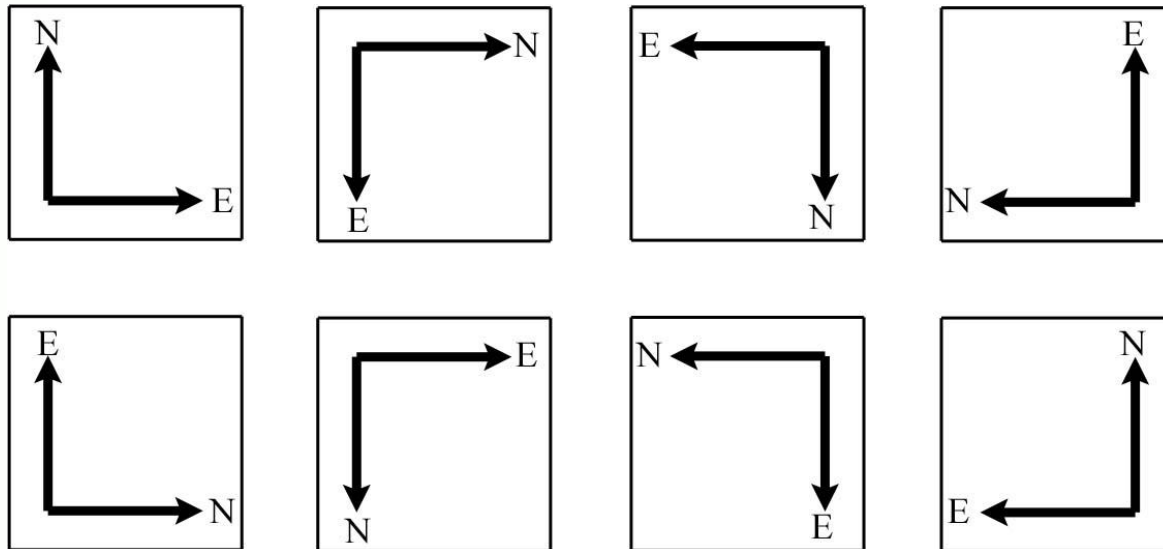
Screen Space

- But it doesn't *have to* be this way. It's only a convention.
- In screen space, for example, $+y$ points down.
- Screen space is how you measure on a computer screen, with the origin at the top left corner.



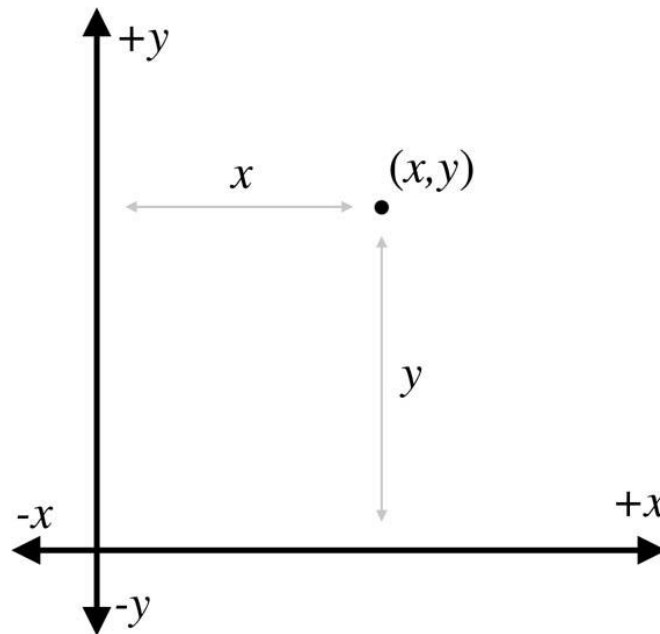
Axis Orientation

There are 8 possible ways of orienting the Cartesian axes.

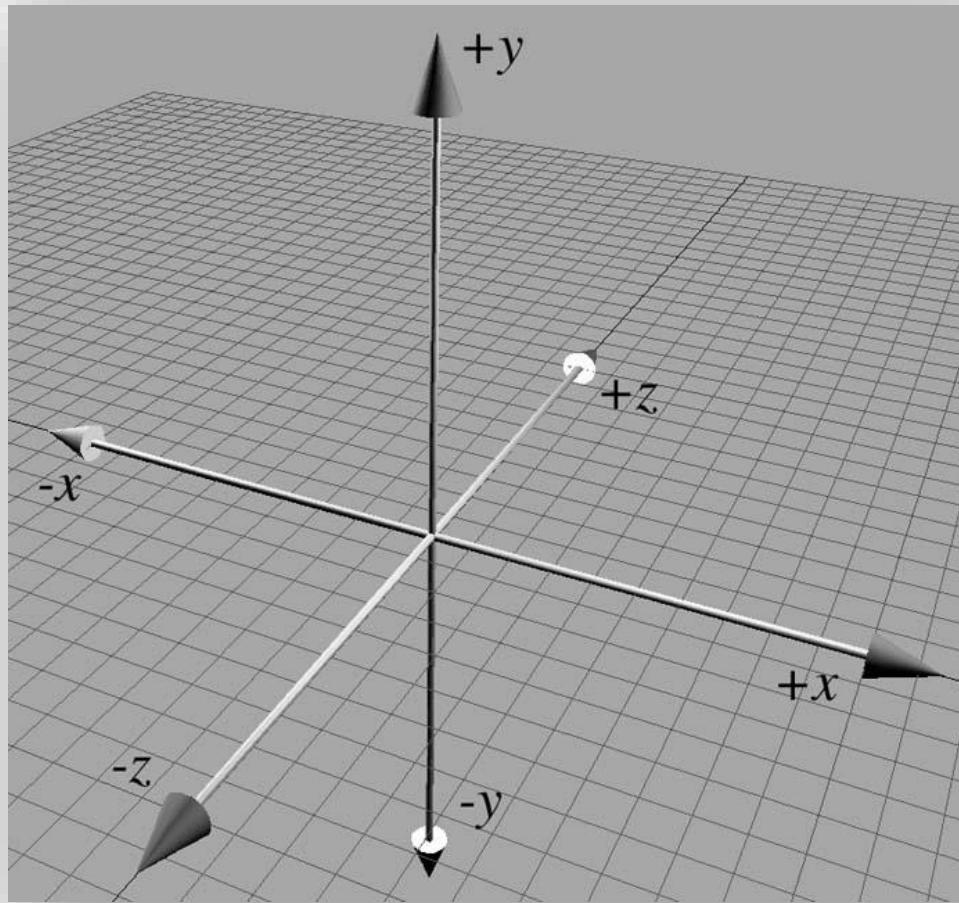


Locating Points in 2D

Point (x,y) is located x units across and y units up from the origin.

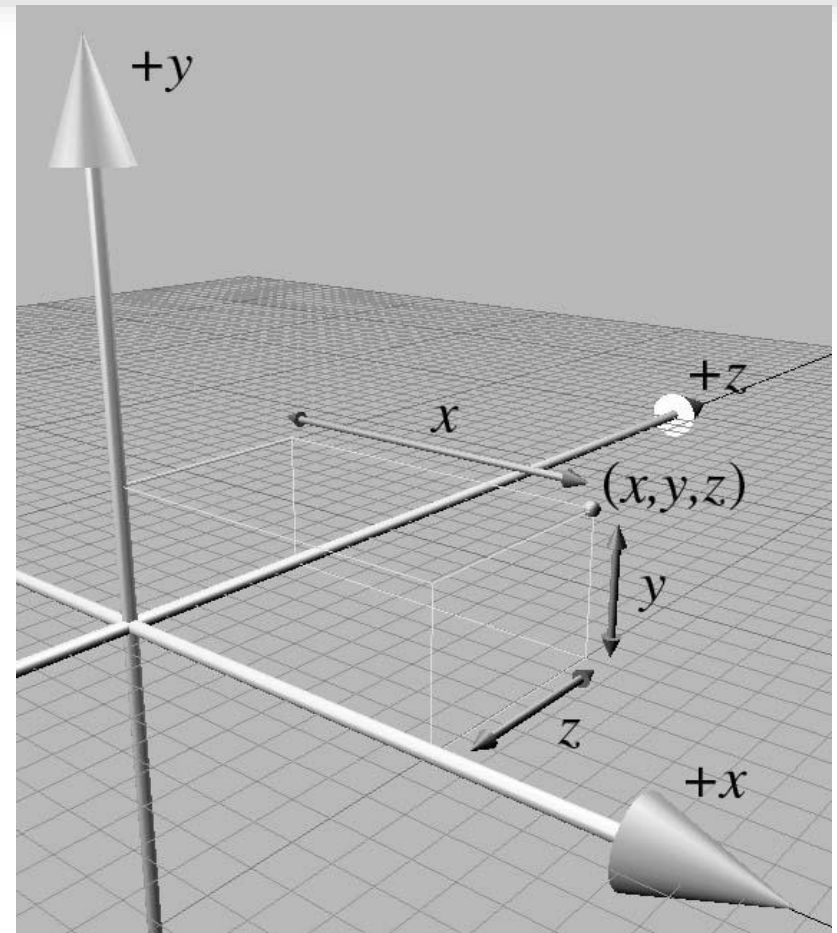


3D Cartesian Space



Locating Points in 3D

Point (x,y,z) is located x units along the x -axis, y units along the y -axis, and z units along the z -axis from the origin.



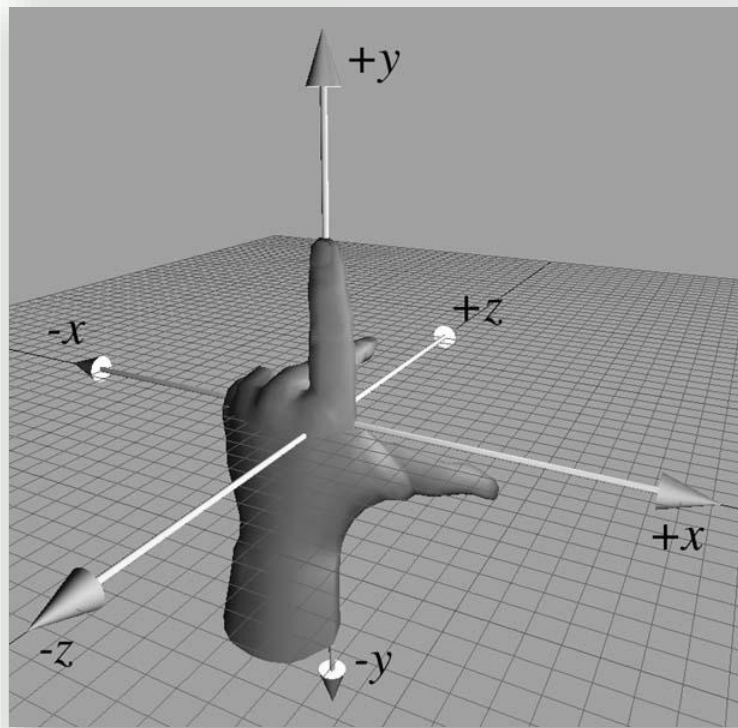
Axis Equivalence in 3D

- Recall that in 2D the alternatives for axis orientation can be obtained by rotating the map around 2 axes.
- As we said, this is not true of 3D coordinate space.

Visualizing 3D Space

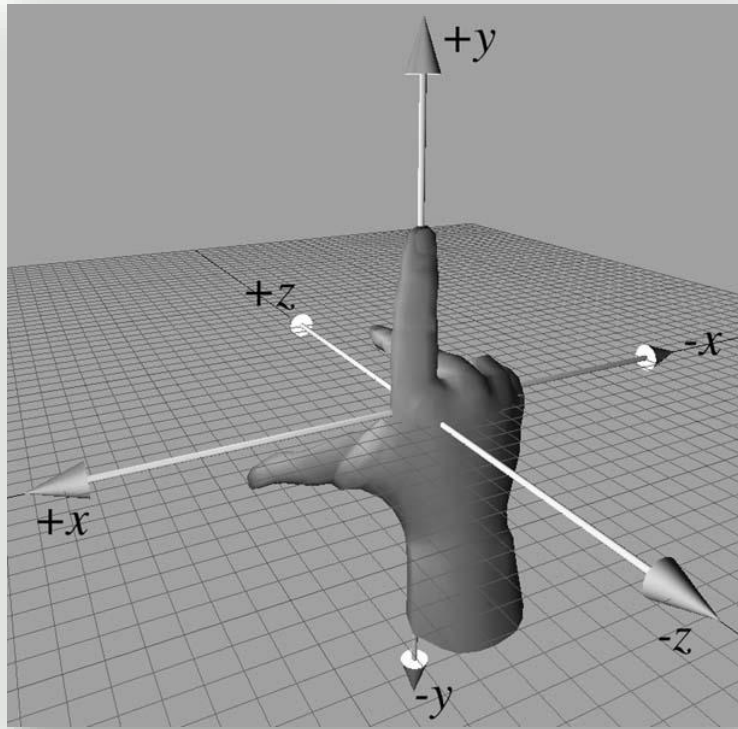
- The usual convention is that the x -axis is horizontal and positive is right, and that the y -axis is vertical and positive is up
- The z -axis is depth, but should the positive direction go forwards “into” the screen or backwards “out from” the screen?

Left-handed Coordinates



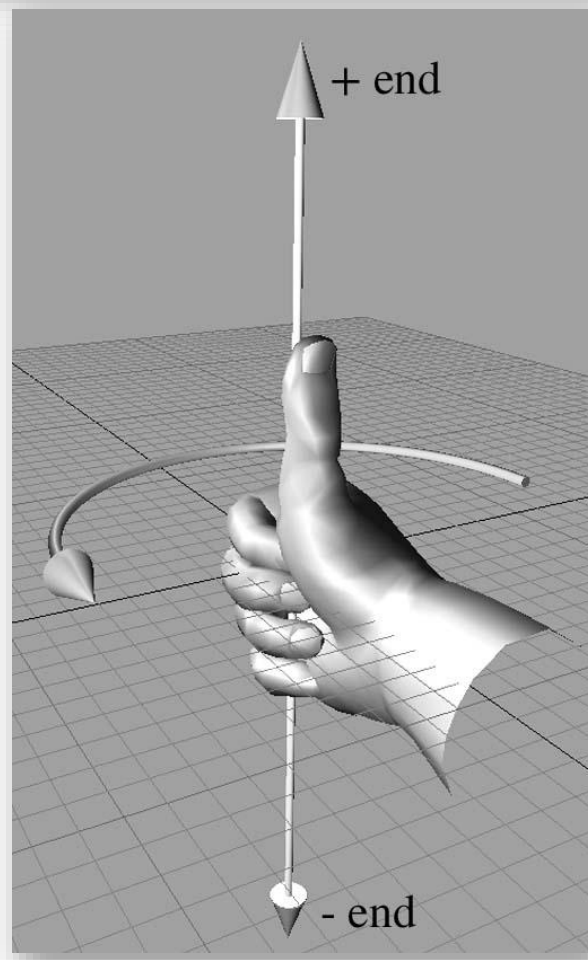
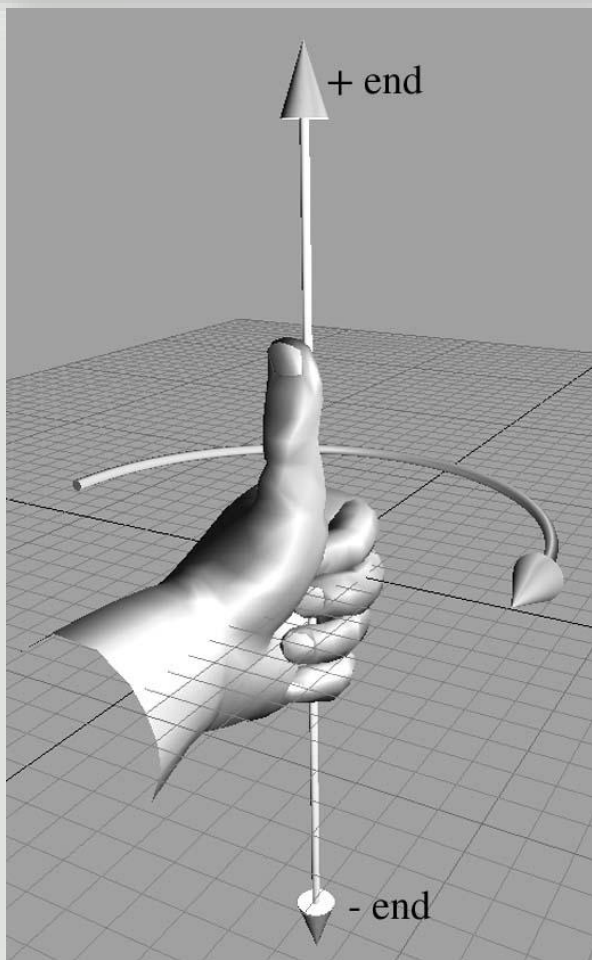
- $+z$ goes “into” screen
- Use your left hand
- Thumb is $+x$
- Index finger is $+y$
- Second finger is $+z$

Right-handed Coordinates



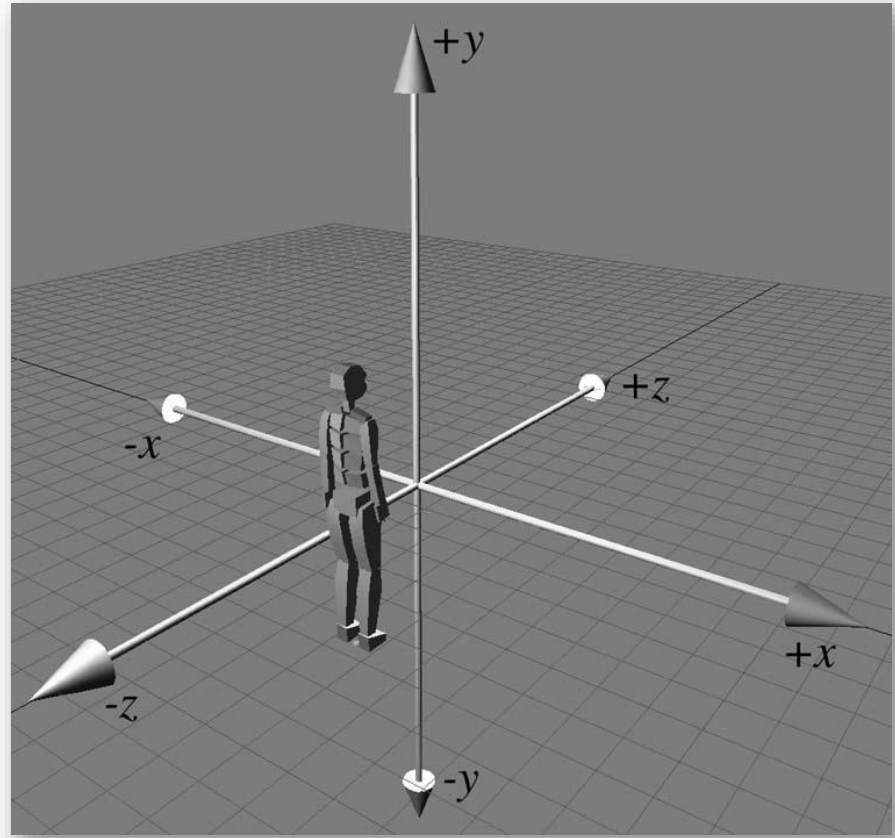
- $+z$ goes “out from” screen
- Use your right hand
- Thumb is $+x$
- Index finger is $+y$
- Second finger is $+z$
- (Same fingers, different hand)

Positive Rotation

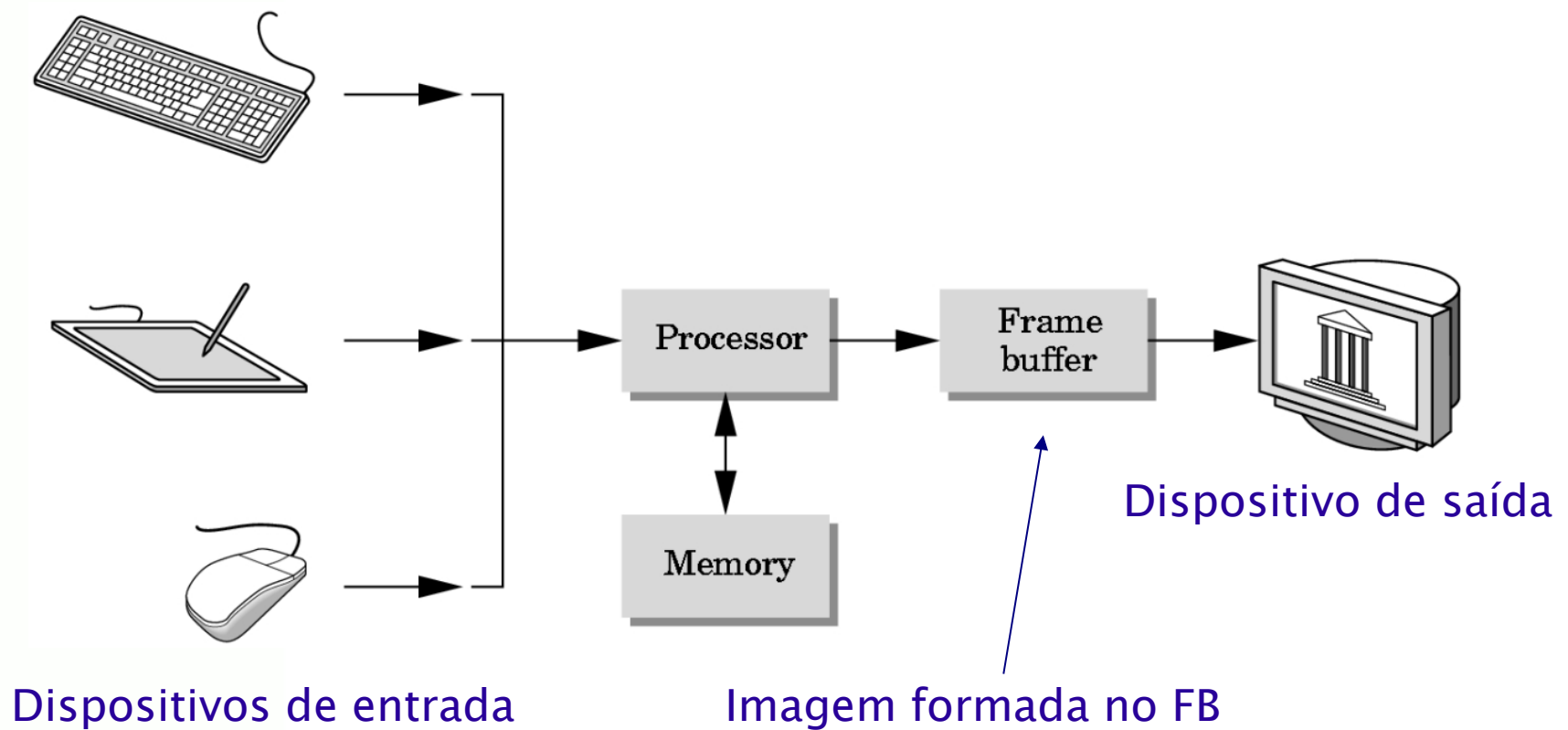


Our Convention

For the remainder of these lecture notes, as in the book, we will use a left-handed coordinate system.



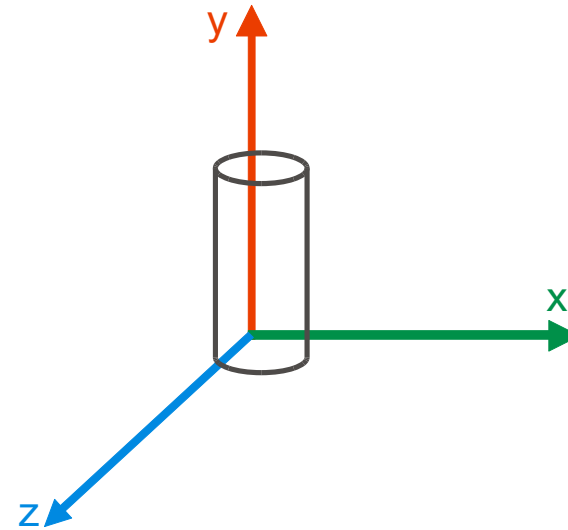
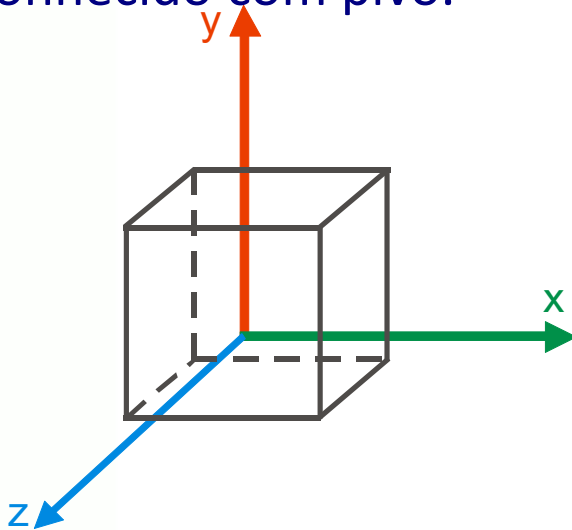
Sistema Gráfico Básico



Sistemas de Coordenadas

- Sistema de Referência do Objeto

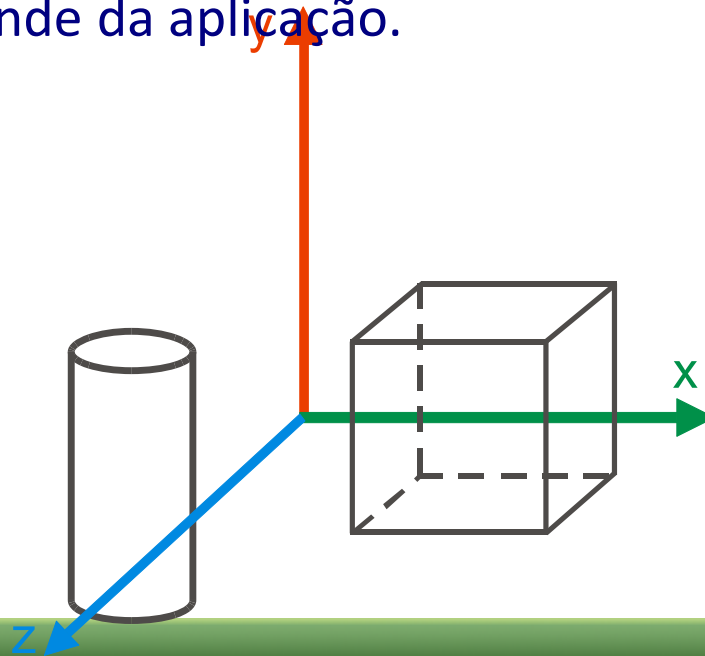
- Cada objeto seja um mini-universo individual.
- Muitas vezes o centro do sistema de coordenadas coincide com o seu centro de gravidade.
- Conhecido com pivô.



Sistemas de Coordenadas

- Sistema de Referência do Universo

- É usado para descrever os objetos em termos das coordenadas utilizadas pelo usuário.
- A escala pode ser em metros, centímetros, milímetros ou outra medida, depende da aplicação.



Sistemas de Coordenadas

- Sistema de Referência Normalizado
 - $0 \leq x \leq 1$ e $0 \leq y \leq 1$, sendo x e y coordenadas horizontais e verticais possíveis.
 - Tornar a geração das imagens independente do dispositivo.
- Sistema de Referência do Dispositivo
 - Utiliza coordenadas que podem ser fornecidas diretamente para um dispositivo de saída específico.
 - Nos equipamentos, depende geralmente da resolução possível e da configuração definida pelo usuário.

Sistemas de Coordenadas

